Simulation vs Practical Development of an RC Aircraft

1. Aim and assumptions

The aircraft has to be developed for a national aeromodelling competition with limited time, budget and workshop access. I would use a **hybrid approach**: simulation to compare possible designs before committing to a build, followed by an early prototype and measured tests. Simulation alone cannot account for every manufacturing or flight issue, while building repeatedly without analysis can waste materials and time.

Assumptions: a conventional fixed-wing RC aircraft; the competition rules define the mission and limits for size, mass and propulsion; basic CAD and aerodynamic tools are available; and flight tests can be carried out in a permitted area with appropriate supervision. No physical test data are available for this proposal, so performance values are not presented as measured results.

2. Simulation-led and prototype-led design

Simulation-led design is useful for screening wing layouts, estimating lift and drag trends, and checking whether a proposed configuration is worth building. XFLR5 can support preliminary low-speed wing and airfoil analysis; OpenVSP can define aircraft geometry. These results depend on input quality and model assumptions. They should be treated as estimates, not proof that the finished aircraft will fly as predicted.

Prototype-led design exposes issues that are difficult to represent accurately in a simple model: actual mass, loose or binding linkages, motor and propeller behaviour, structural flex, and handling. Its disadvantage is that each build or repair costs time and materials. Flight results can also be misleading if wind, battery condition or setup changes between tests.

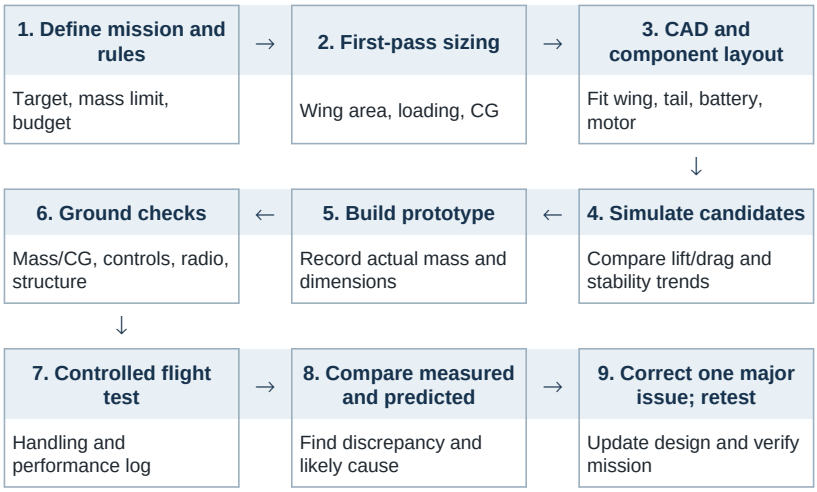
Comparison

Criterion	Simulation-led	Prototype-led
Cost	Low cost for trying many virtual options; software and computing still have a cost.	Materials, damaged parts and repairs make repeated iterations more expensive.
Time	Fast for initial comparisons; detailed analysis can take time.	A simple test may be quick, but building and repairing take time.
Accuracy	Limited by assumptions, inputs and the method used.	Measures the real aircraft, but instruments and test conditions introduce error.
Iteration speed	Geometry and parameters can be changed quickly.	Changes may require rebuilding, rebalancing or repairs.
Failure detection	Finds only problems represented by the model.	Reveals integration, propulsion, control and handling problems.
Final performance	Helps choose a promising design; cannot confirm real performance.	Checks actual performance in the intended configuration.

Decision: use simulation to reduce the number of weak candidates, but let measured results decide whether the aircraft meets the mission. Build the first credible design early enough to leave time for corrections.

3. Development workflow

The process should be iterative. If a test disagrees with the prediction, first identify the likely cause rather than changing several design features at once.



4. Parameters to study

Simulation parameter	Purpose / output
Lift vs angle of attack	Estimate the useful operating range and likely stall region.
Drag polar and lift-to-drag ratio	Compare aerodynamic efficiency between candidate layouts.
CG and longitudinal stability	Check how mass distribution affects trim and pitch stability.
Wing area and aspect ratio	Compare lift and induced-drag trends for the mission.
Structural deflection	Estimate whether the structure may flex excessively under expected loads.

Physical check	Method and reason
Mass and CG	Use a scale and balance point; confirms the built aircraft matches the design assumptions.
Static thrust, current and voltage	Measure with suitable instruments and a secure test setup; checks propulsion output and electrical demand.
Take-off distance and handling	Record under controlled conditions; checks whether predicted performance is achieved in practice.
Control response	Ground checks and flight observations; reveals binding, incorrect direction or weak response.

5. Validating the model

A fair comparison needs the same aircraft configuration and reasonably matched conditions. Before testing, save the model's geometry, mass, CG, assumed flight speed and predicted outputs. After building, measure the actual dimensions and mass, record battery condition and weather, and note instrument precision. Compare like with like—for example, measured static thrust with the corresponding propulsion estimate, rather than comparing unrelated values.

For a non-zero measured value, percentage error can be calculated as:

> **Percentage error** = |simulated value – measured value| / |measured value| × 100%

Keep the signed difference as well, because it shows whether the model tends to overpredict or underpredict. If the gap is large, check input errors, wind, measurement uncertainty, propeller effects, structural flex and control behaviour. Update the model only when there is a defensible reason, then check it against a separate test or operating condition. Matching one result by adjusting arbitrary inputs is not sufficient validation.

6. If the simulated aircraft performs well but the prototype will not take off

Treat this as fault isolation. Do not immediately increase power or alter several features together. Start with simple checks and record what is found.

Possible cause	Check	Next step
Mass higher than model	Weigh the aircraft; compare actual wing area and loading with the model.	Correct the mass record and reassess performance margin.
CG or geometry error	Check balance point, wing alignment, incidence, twist and visible damage.	Correct the setup before further testing.
Propulsion underperforming	Check battery state, wiring, motor operation and measured thrust/current.	Compare measurements with component limits and design requirements.
Mechanical or control issue	Inspect wheels, linkages, servo travel, control direction and binding.	Repair and repeat ground checks.
Conditions differ from model	Review wind, surface, air density and assumed take-off speed.	Update assumptions and repeat only when conditions are suitable.

7. Four-week plan

Week	Work	Output
1 — Design	Confirm mission and rules; do first-pass sizing, CAD and initial simulation.	Requirements, calculations and candidate layout.
2 — Build	Build the first prototype; measure mass/CG; inspect structure, controls and propulsion on the ground.	Prototype and ground-check record.
3 — Test	Carry out controlled tests; log results and compare them with predictions. Change one major variable at a time.	Test log and revised design.
4 — Verify	Fix faults, repeat relevant checks and document final configuration.	Final report and evidence.

Reserve about two days in Week 4 for repairs, weather delays or repeat measurements. Safety checks take priority over the schedule; unresolved faults are not a reason to fly.

Conclusion

For this project, simulation-first and prototype-first are not competing choices. Simulation is best used to narrow the design space; the prototype shows whether the assumptions survive contact with the real aircraft. The final design should be selected from repeatable measurements and documented checks, not from a promising simulation alone.

References

1. XFLR5, General Description and Documentation: https://www.xflr5.tech/xflr5.html
2. NASA, OpenVSP Ground School: https://www.nasa.gov/software/openvsp-ground-school/
3. OpenVSP project repository: https://github.com/OpenVSP/OpenVSP
4. Project X, Problem 3 — Simulation vs Practical Development: https://github.com/b26418-code/Project-X-Selection-Tasks/blob/main/RC-Plane/03_simulation_vs_practical.md